

Maps, charts, abstraction

CMS.100

Agenda

- Reading check-in: D'Ignazio & Klein (10 min)
- Protocol registration (30 min)
- Mini-lecture on maps (10 min)
- Map activity (20 min)
- A3.1 instructions (10 min)

Hand back: reading check-ins from last week

D'Ignazio and Klein: reading check-in

Pick one:

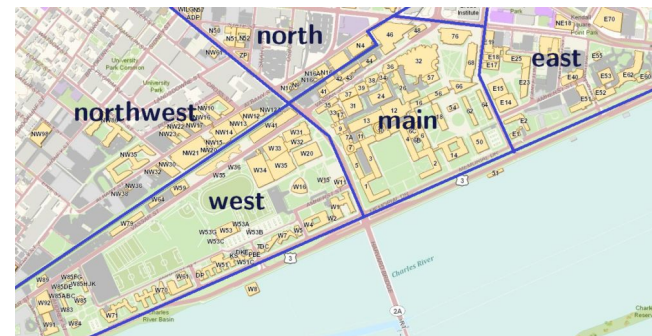
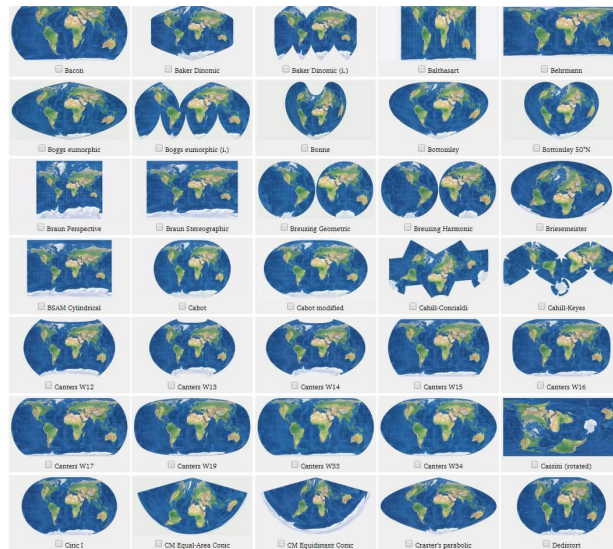
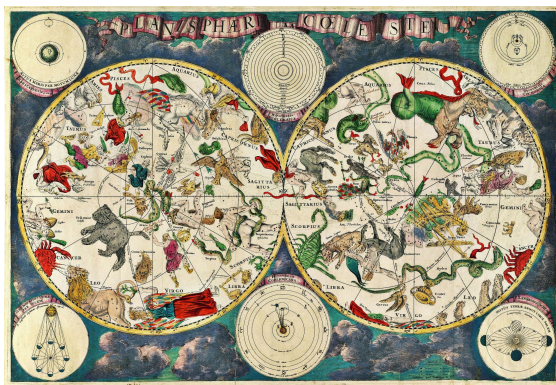
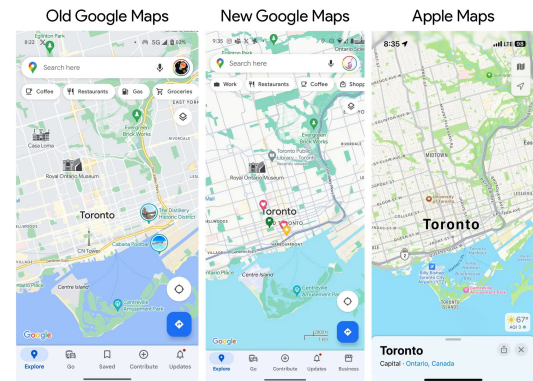
- What does the phrase “data may be useful, but they are not neutral” mean in the context of this reading? Give one example.
- What is “counterdata?” Give one example.

Dataset collection protocol

- **Data set:** What is your data set about? What gap is it filling?
- **Unit of observation:** what is one row of data? What does each number or observation represent? What kind of data is it (e.g., continuous, categorical, etc)?
- **Inclusion criteria:** what belongs in the dataset?
- **Exclusion criteria:** what would be excluded from the dataset?
- **Measuring instrument:** how are you collecting the data? With what?
- **Scope:** how do you decide when a dataset starts and ends? What is the schedule or duration of your dataset?
- **Missing data:** how are you dealing with missing, blank, null, etc. data? What does a blank, NA, or 0 mean in the context of your data?
- **Storage:** file format? File name convention?
- **Claims:** what kinds of arguments could you make with this data?

What makes a map?

Geographic maps, mental maps, thematic maps, and maps of non-terrestrial spaces...



What is a map?

“Maps have three basic attributes: **scale, projection, and symbolization**. Each element is a source of distortion.”

— Mark Monmonier, *How to Lie with Maps*

“a symbolic representation of **selected characteristics of a place**, usually drawn on a flat surface.”

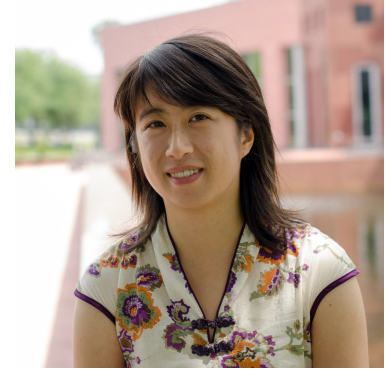
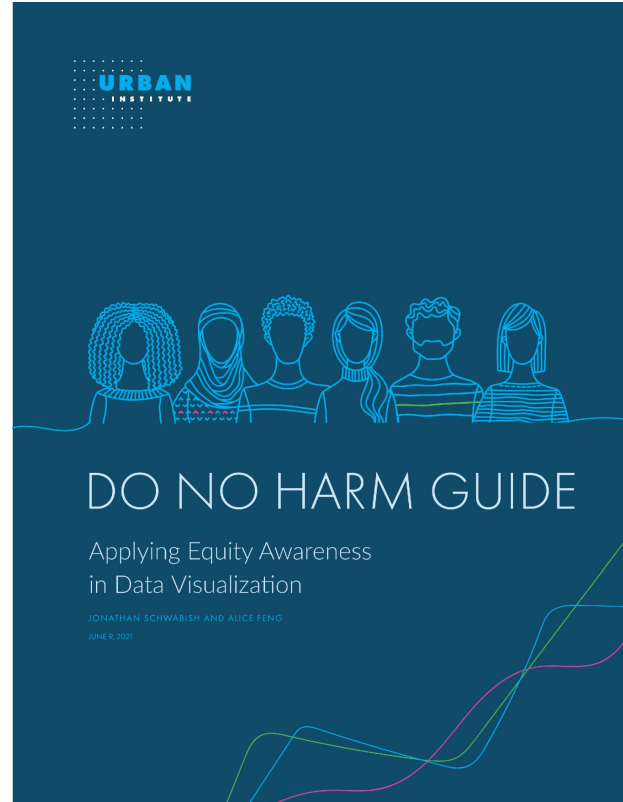
— *National Geographic*

“The map is currently defined as follows: ‘A map is a **symbolized representation of geographic reality**, representing selected features or characteristics, resulting from the creative effort of its author’s execution of choices, and is designed for use when spatial relationships are of primary relevance.’”

— International Cartographic Association

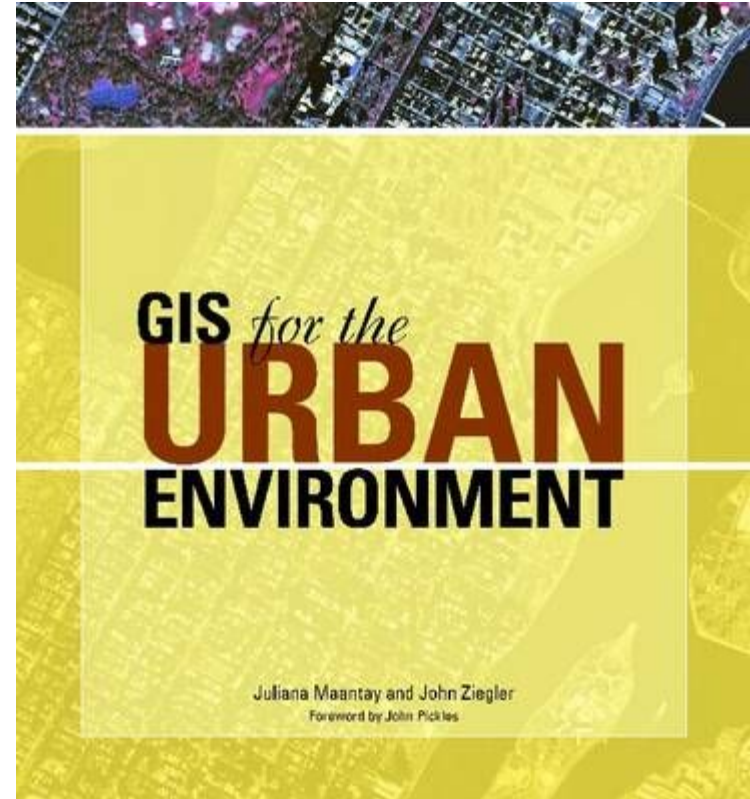
Mapping...

- Illustrates an author's **point of view**
- as a form of **visual storytelling**
- is an **instrument** of power (maps can reinforce control or be tools for change)
- **embeds values** in how data is represented



What is a map?

- **Different scales:** sub-microscopic, human-scale, macro-scale)
- **Tangible or intangible** (e.g. population)
- **Real or fictional:** projected future scenarios
- **Temporal:** represent one moment in time, a duration of time, or a change over time



(1) Different scales

Seattle



Seattle



Seattle



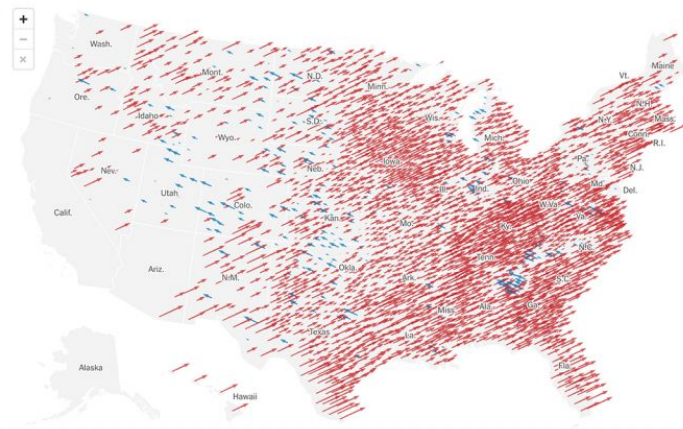
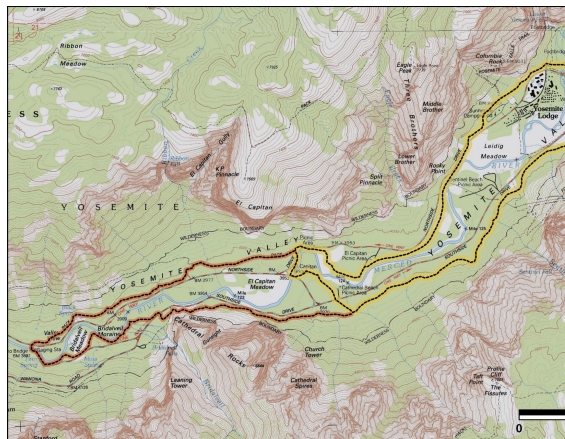
Seattle



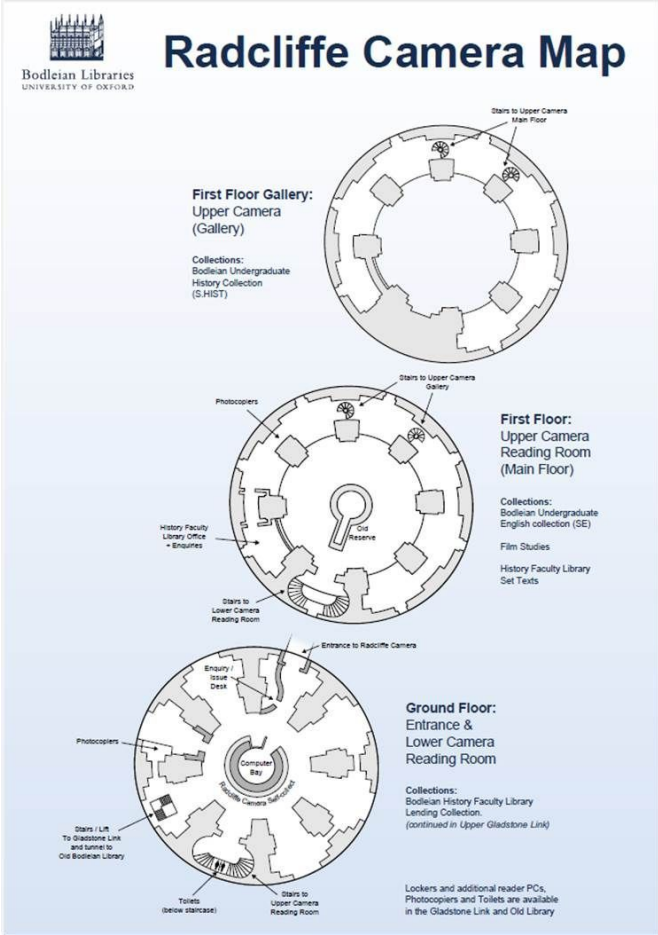
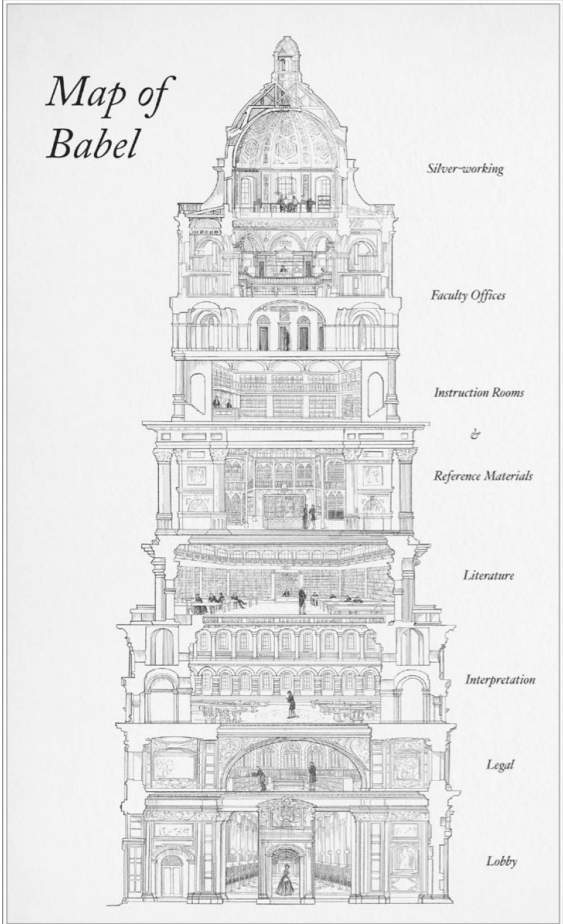
(2) Tangible or intangible

Left, tangible: Yosemite Valley, National Park Service

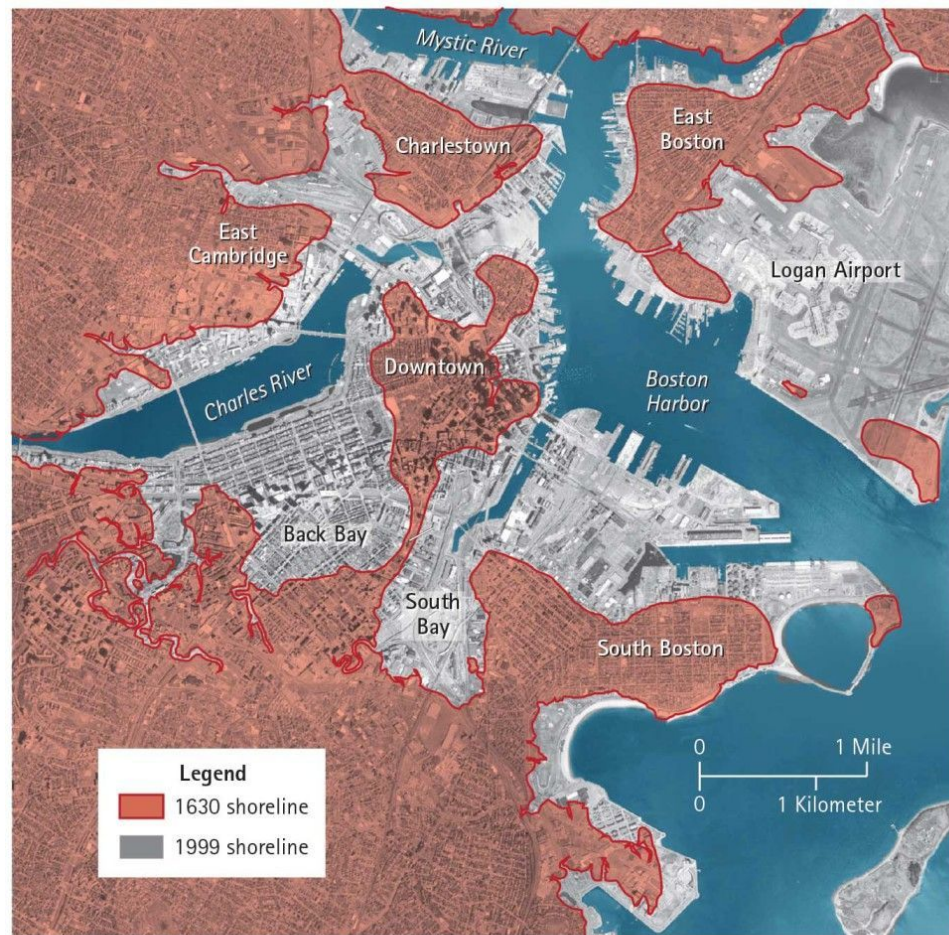
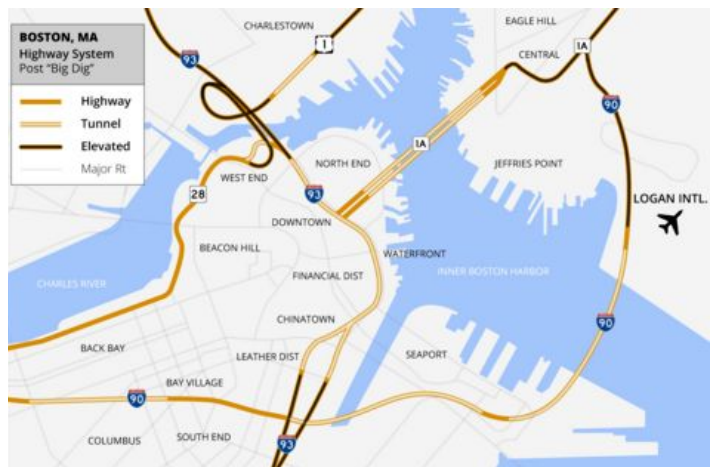
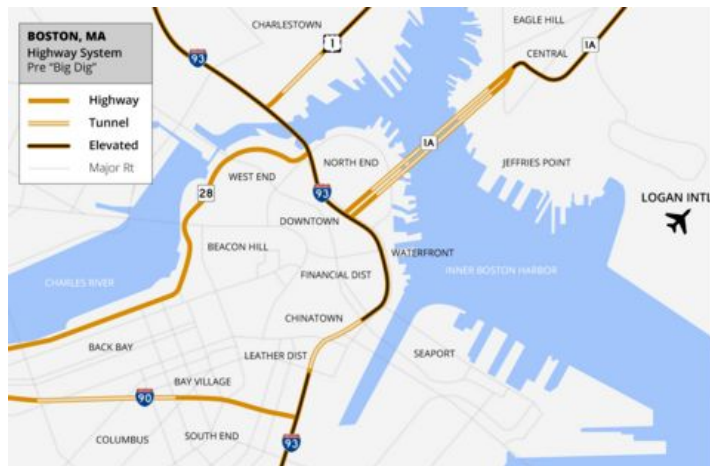
Right, intangible: Election 2020 map, *New York Times*



(3) Real or fictional



(4) Temporal

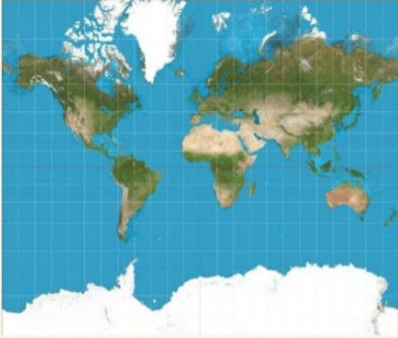


Source: Adapted from Weiskel, Peter K., Lora K. Barlow, and Tomas W. Smieszek. *Water Resources and the Urban Environment, Lower Charles River Watershed, Massachusetts, 1630–2005*. U.S. Department of the Interior, United States Geological Survey, in cooperation with the U.S. Environmental Protection Agency and the Massachusetts Department of Environmental Protection (2005), Circular 1280, Figure 9, p16.



MAP PROJECTIONS

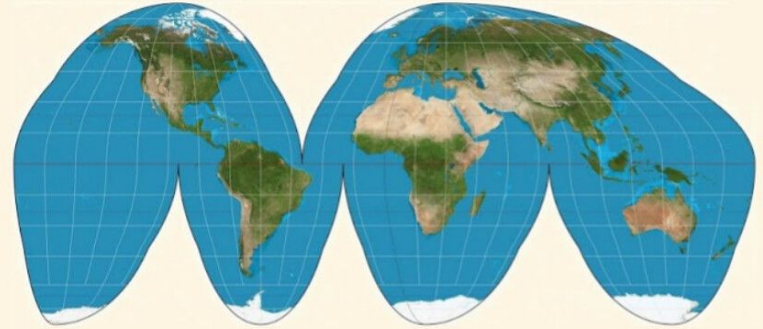
MERCATOR



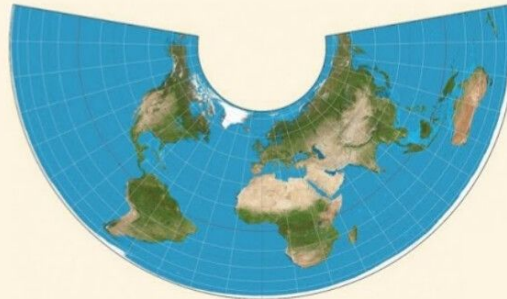
GALL-PETERS



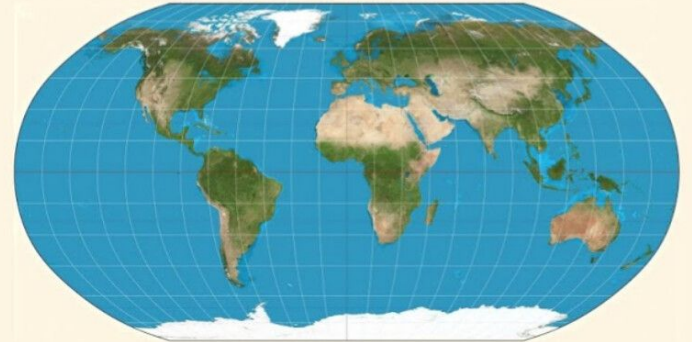
GOODE-HOMOLOGOSINE



WATERMAN BUTTERFLY

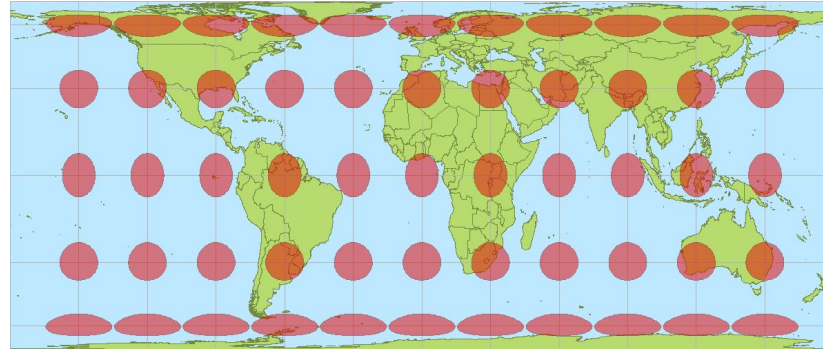
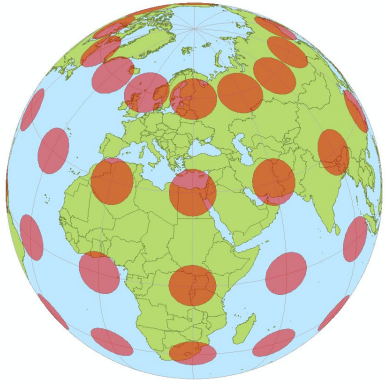


ALBERS

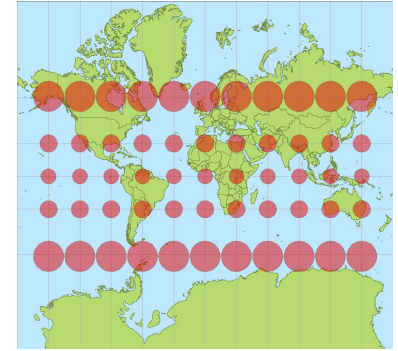


ROBINSON

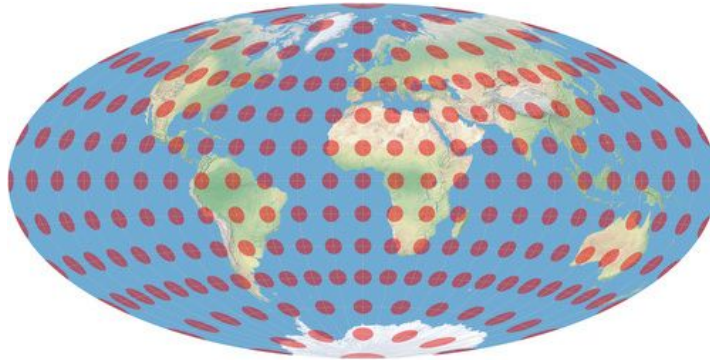
Tissot's Indicatrix (1859)



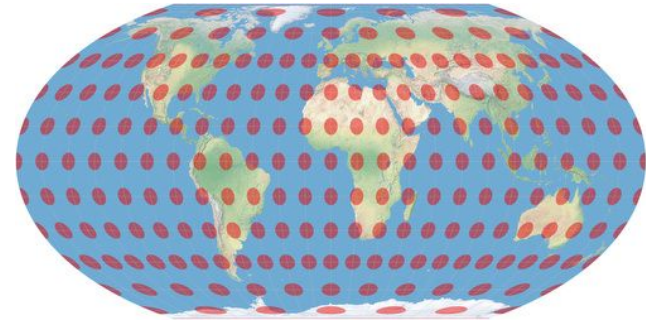
Behrmann



Mercator



Hammer



Wagner

“Matrix of domination”

The four domains of the matrix of domination

Structural domain

Organizes oppression: laws and policies.

Disciplinary domain

Administers and manages oppression.
Implements and enforces laws and policies.

Hegemonic domain

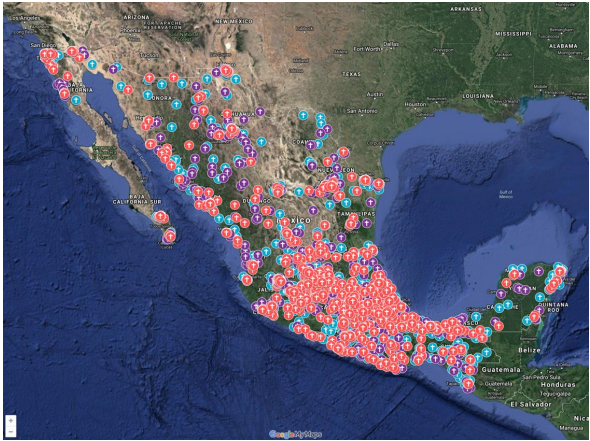
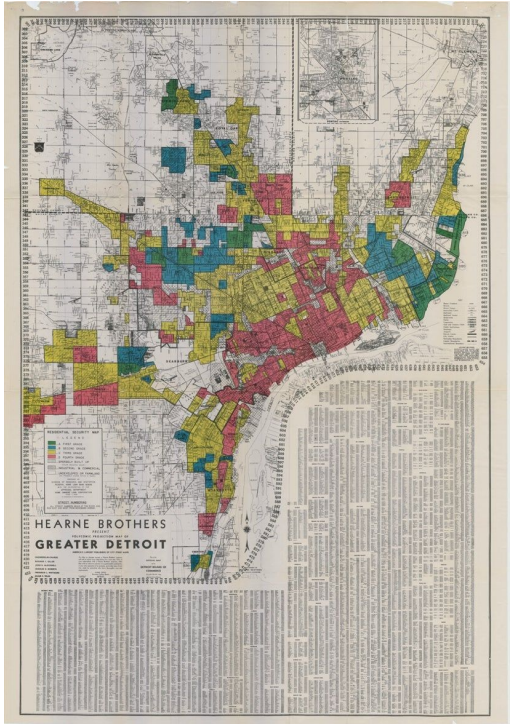
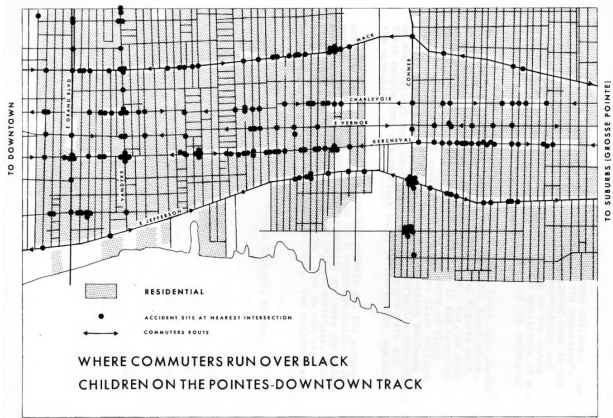
Circulates oppressive ideas: culture and media.

Interpersonal domain

Individual experiences of oppression.

Chart based on concepts introduced by Patricia Hill Collins in *Black Feminist Thought: Knowledge, Consciousness, and the Politics of Empowerment*.

A tale of three maps...



Counter-data collection

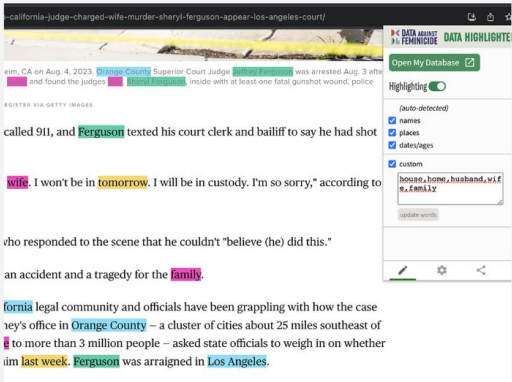


About Tools Publications Community Course Glossary Contact

Search



English



Co-designed Tool #1 Data Highlighter

The **Data Against Feminicide Highlighter** is a plug-in for the Chrome browser. Its function is to highlight words in the text of a web page that are useful for recording cases, specifically names, places, dates, and words selected by the user. The plug-in also allows a database to be opened side-by-side (or top/bottom) simultaneously with the article being analyzed, to facilitate data entry. It also allows the article URL to be shared via email. It works in Spanish, English, and Portuguese and on any web page, but is optimized for news media websites.

Install extension

Installation and user guide

Amienta co-diseñada #2: Alerta de correo electrónico

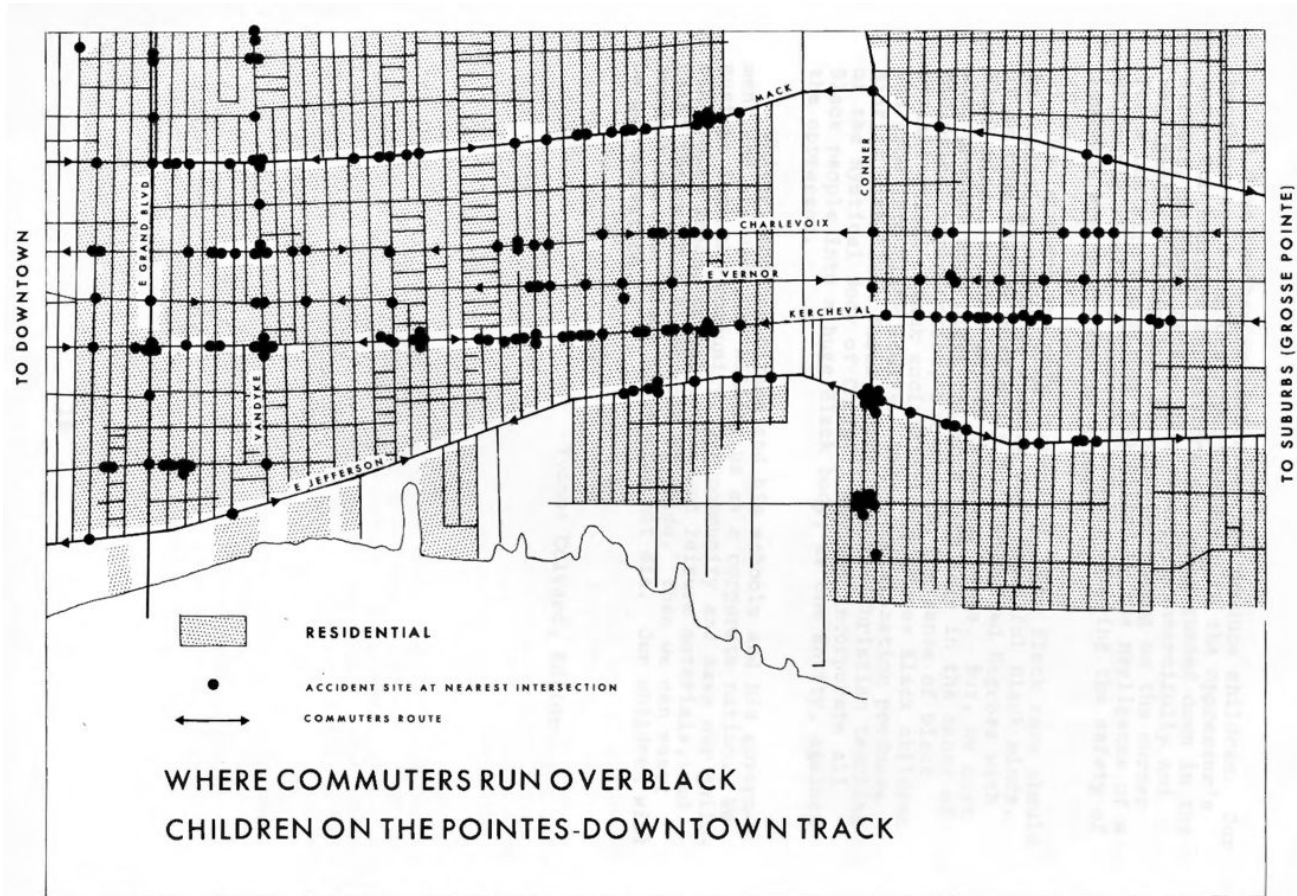


Co-designed Tool #2 Email Alerts

The **Data Against Feminicide Email Alerts** combines a media database, a machine learning algorithm, and email alerts to report potential cases of femicide. The system searches the MediaCloud database for news stories based on search terms and the region determined for each project. It then filters the results through a machine learning algorithm specifically trained to calculate the probability that an article refers to a case of femicide and to group articles from different sources that refer to the same case. Finally, an email is sent out with the relevant articles, based on the frequency determined by each user. The MediaCloud system allows users to view the sources and also suggest new relevant sources.

If you or your group are interested in using **Data Against Feminicide's Email Alerts** or in exploring a translation project, please send an email to datoscontrafemicidio@gmail.com.

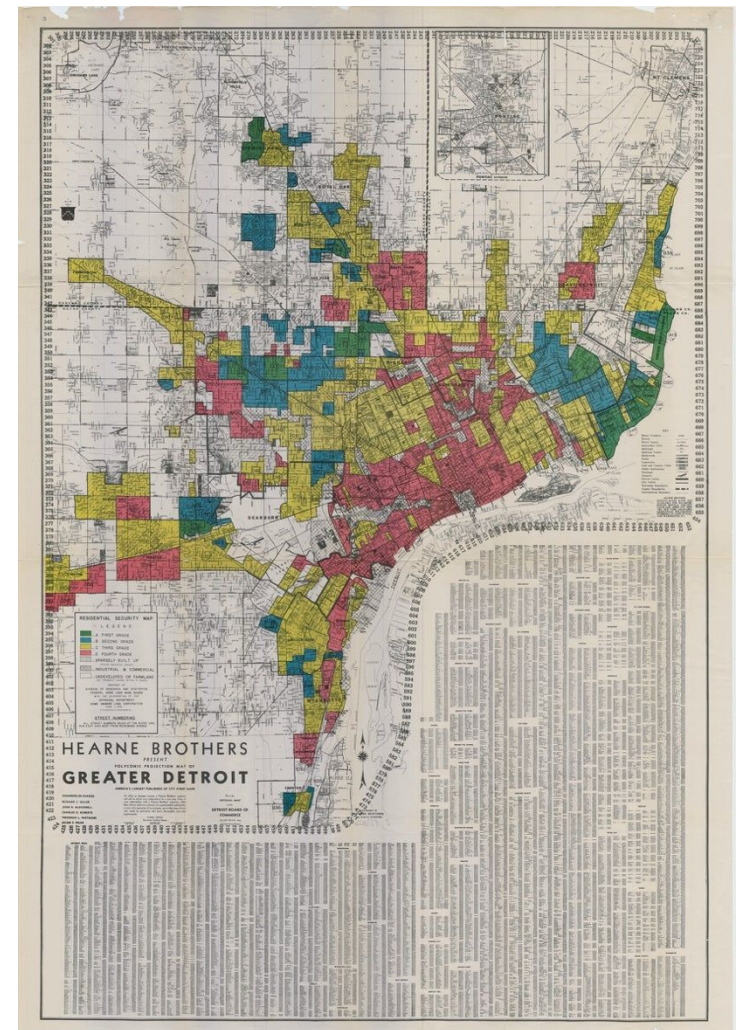
Traffic fatalities in Detroit



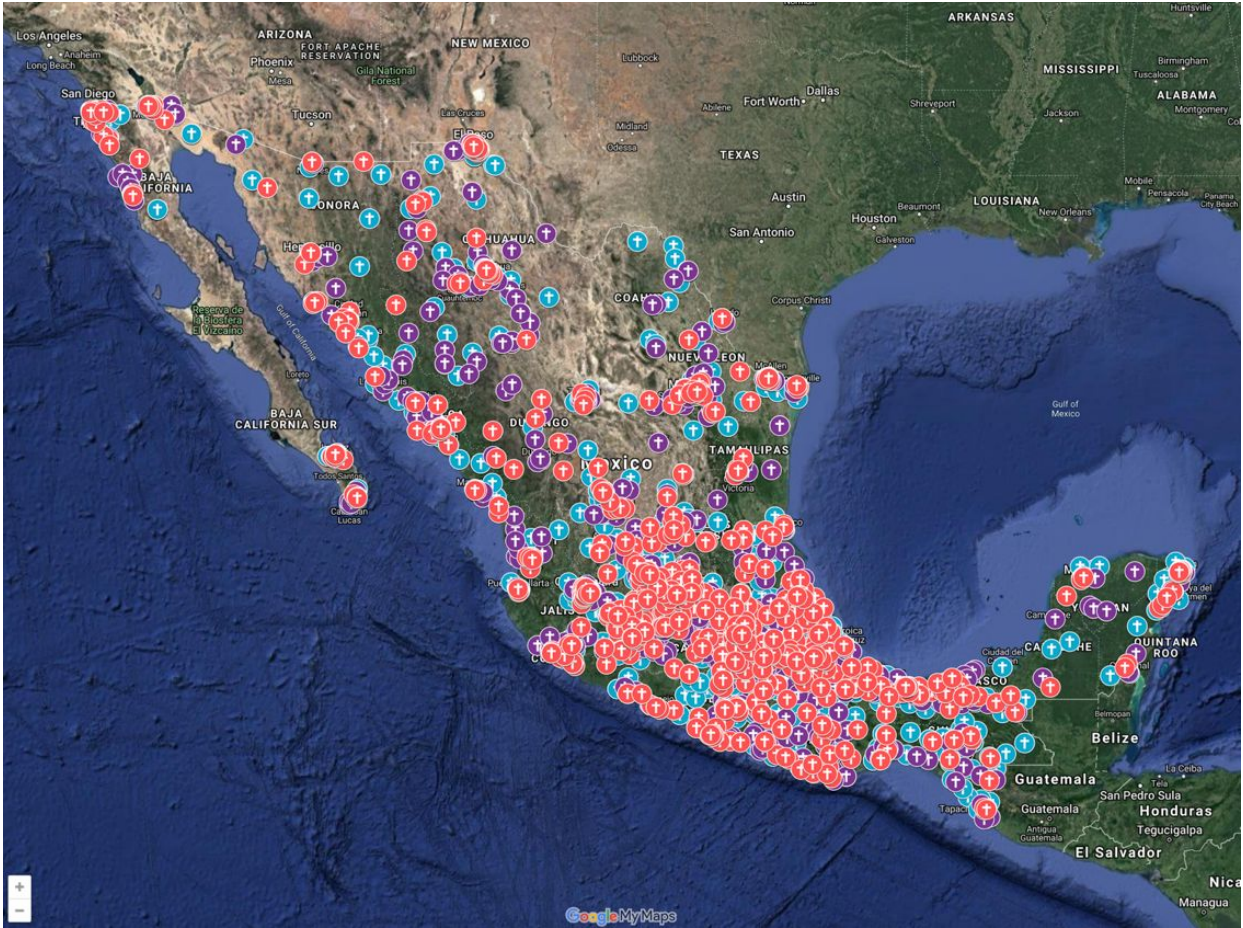
Redlining maps

Redlining was the practice of categorically denying access to mortgages.

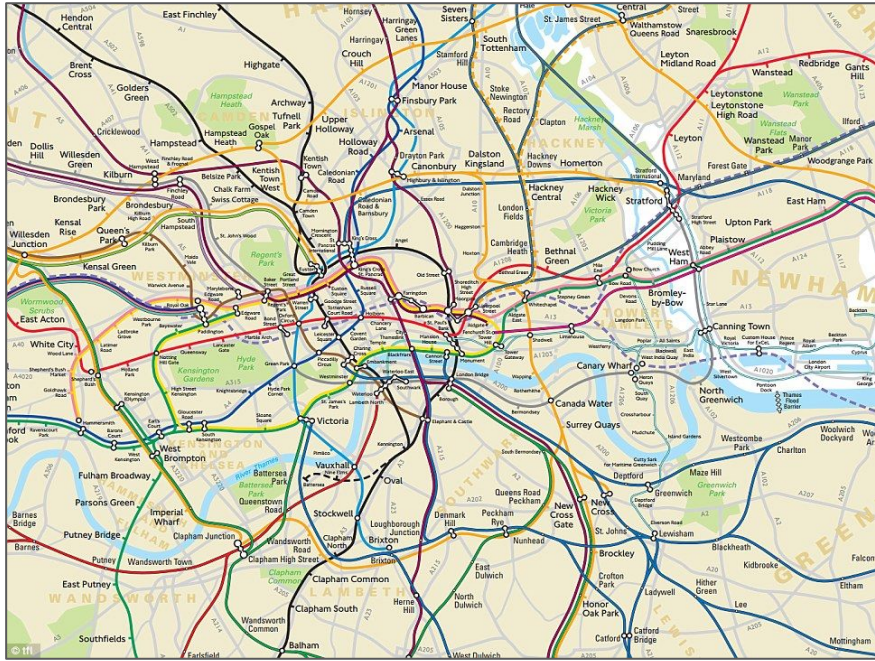
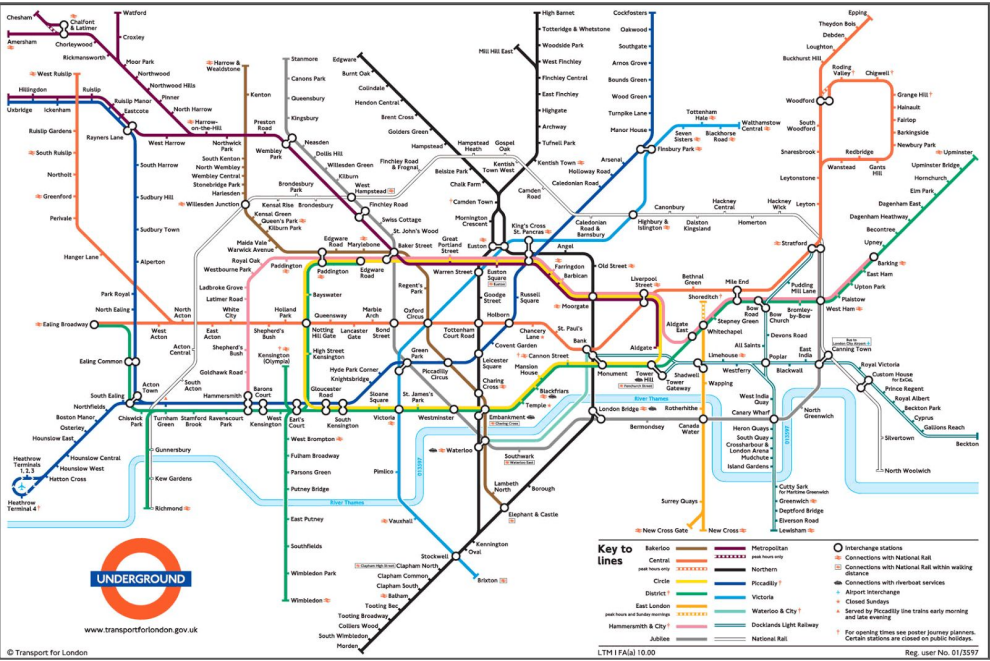
If residents were African Americans or, to a lesser extent, immigrants or Jews, the **Home Owners' Loan Corporation** (government-sponsored corporation created as part of the New Deal, 1933; defunct as of 1954) deemed them a threat to the stability of home values.



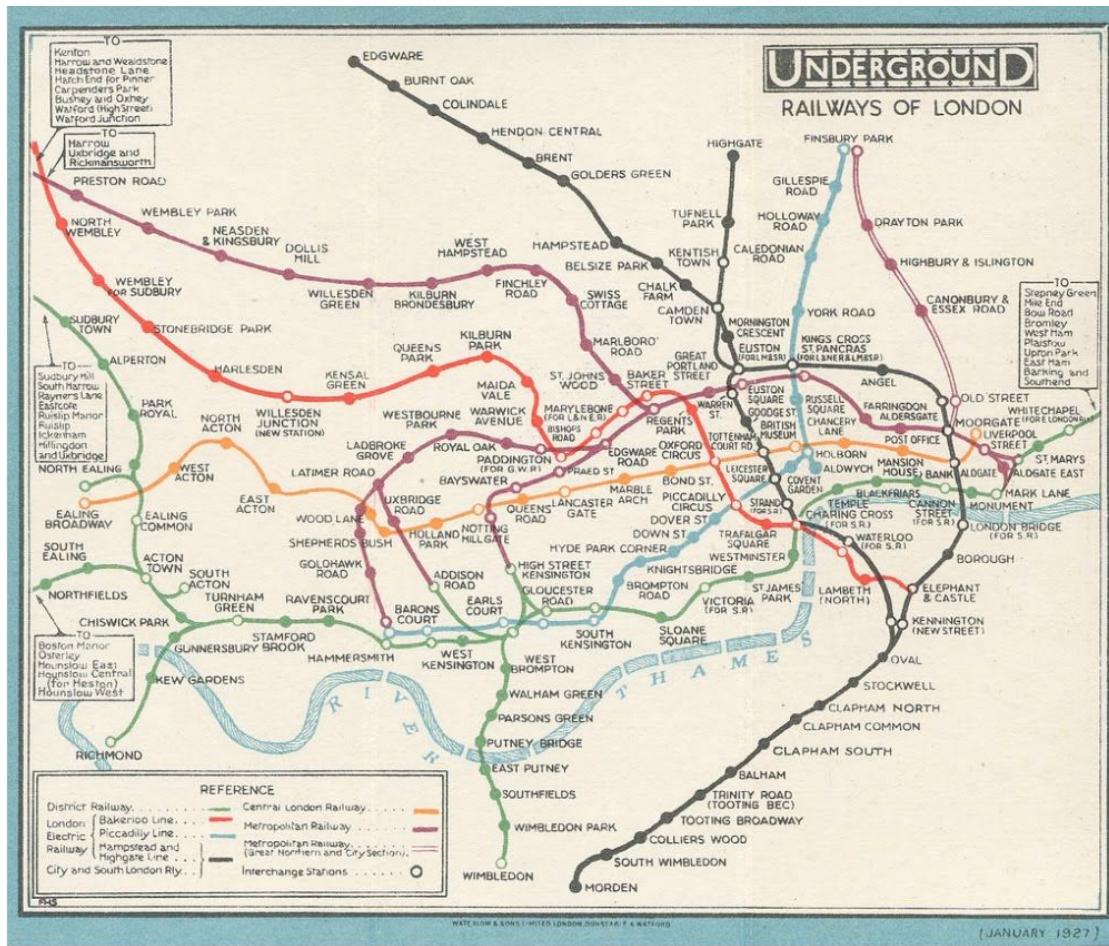
Data Against Feminicide



London Tube map



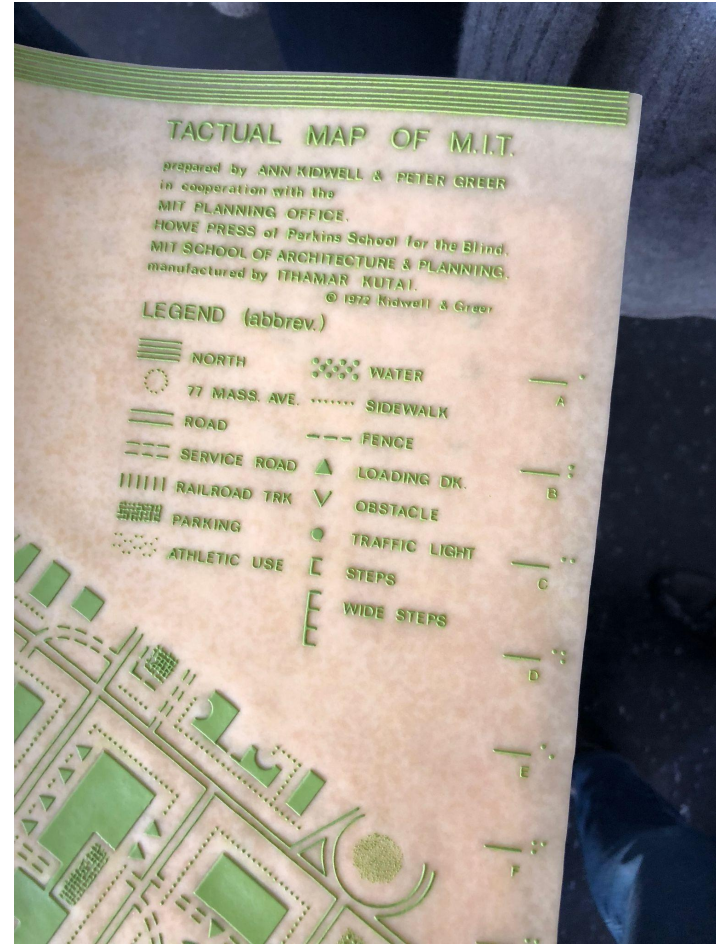
London Tube map (1927)



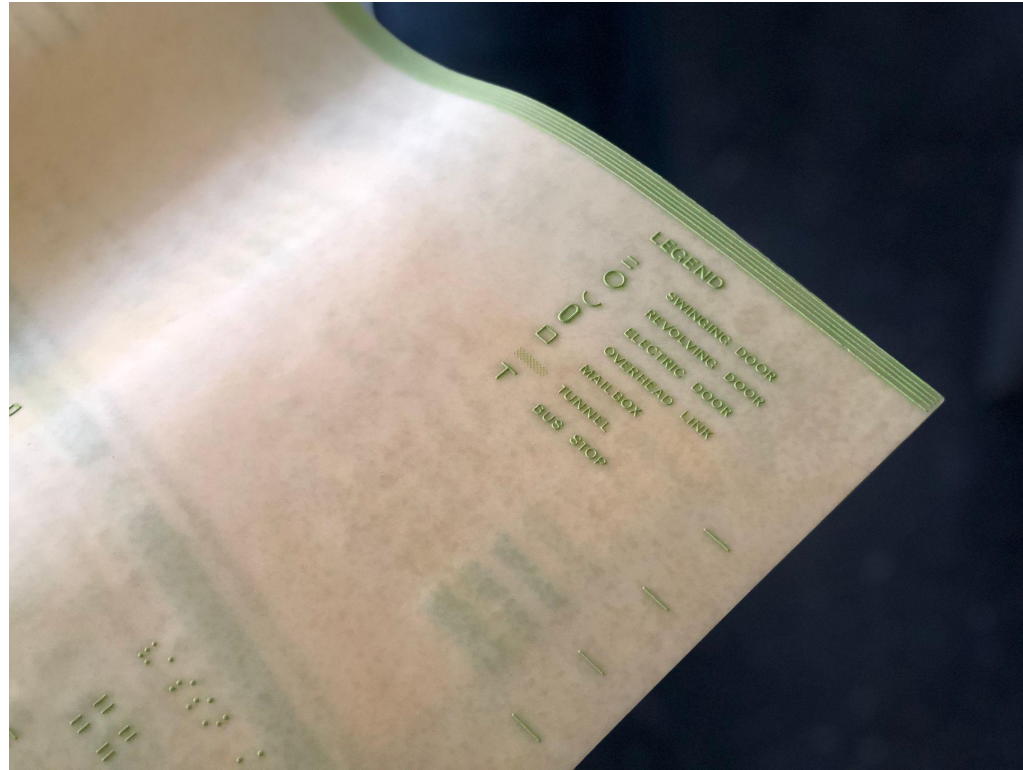
London Tube map (1933)



Tactile map of MIT (1972)



Tactile map of MIT (1972)



Map design fundamentals

- Audience, purpose & design intent (narrative visualization)
- Essential map elements
- Basemap
- Visual Hierarchy
- Color

Interactive Web Map Features

- Map extent / zoom level
- Interaction



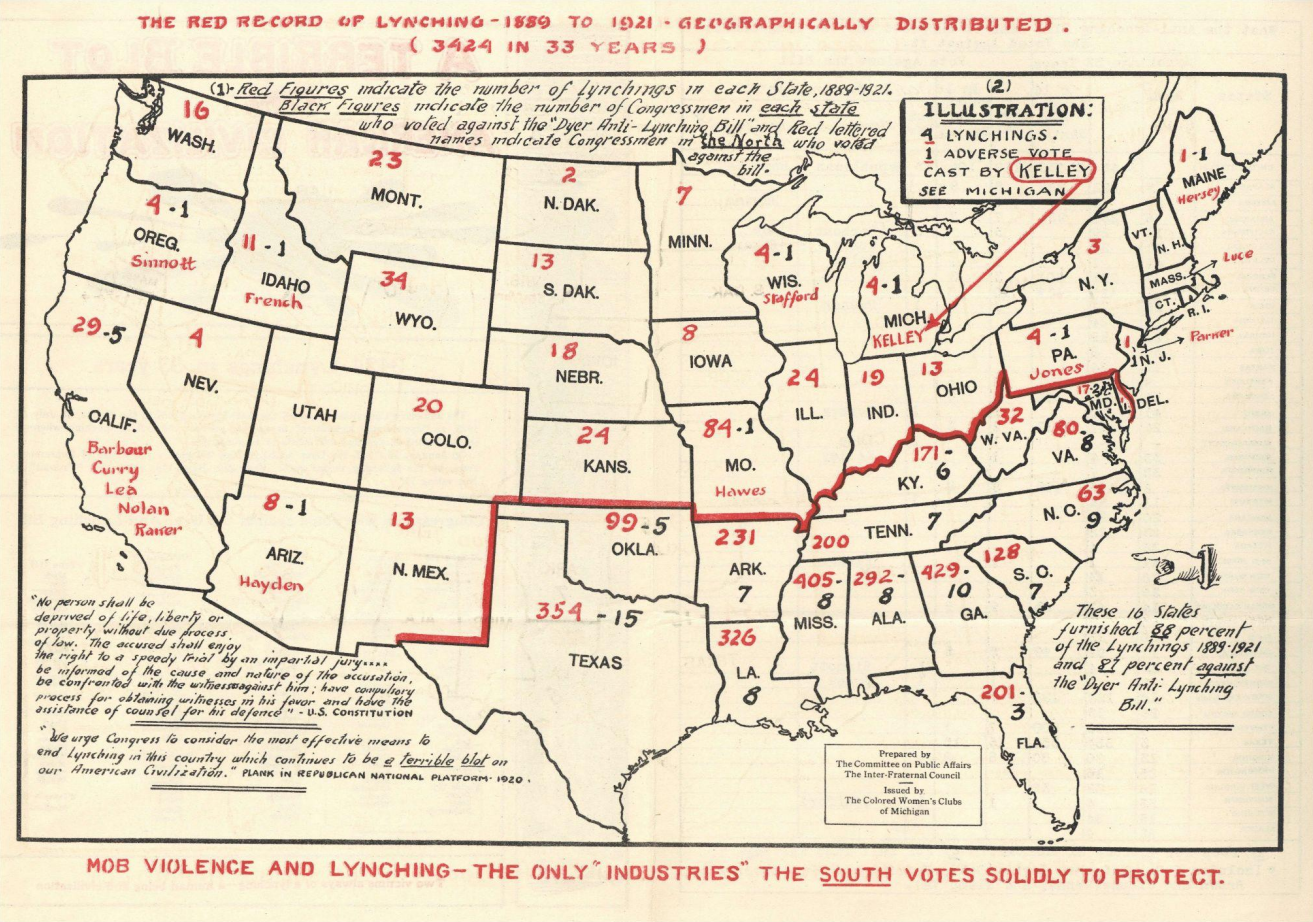
Assignment 3.1 (1 slide)

Due: Oct 7 in class, add slide to shared deck (presentations run Oct 7 *and* Oct 9)

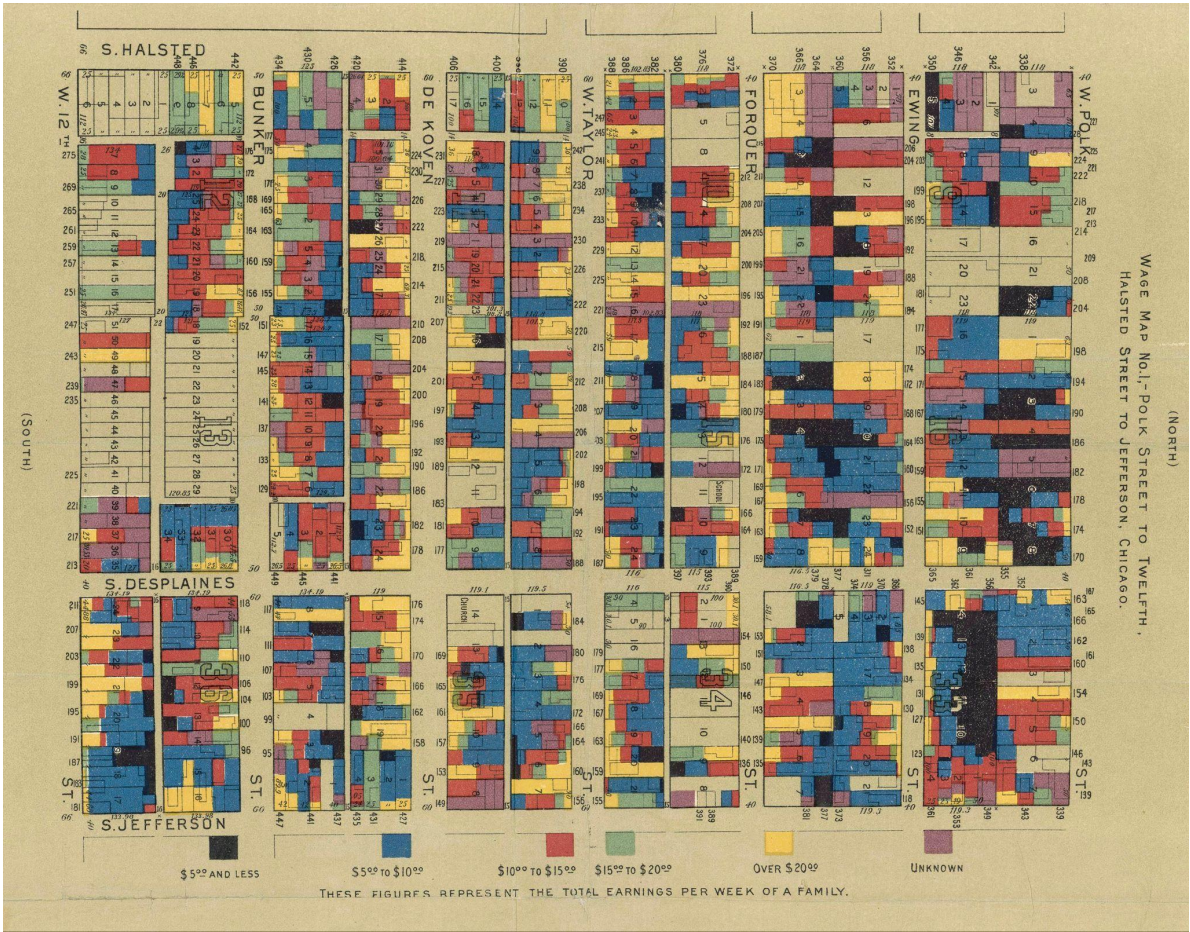
Add a slide to the shared deck with no more than 3 short ideas on what you would like to research for your final project (1–3 sentences per idea).

Be prepared to speak for ~3 minutes and answer questions.

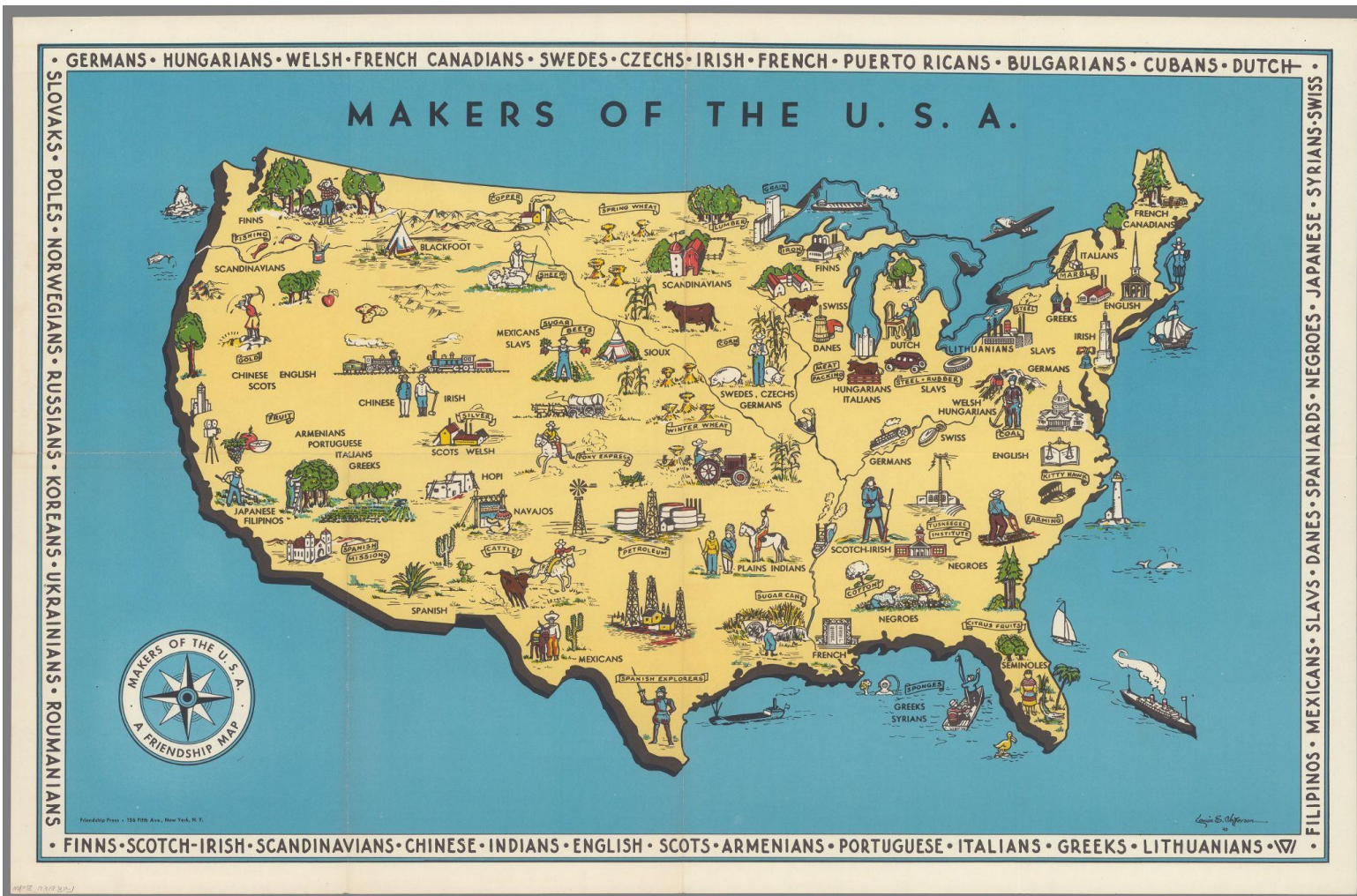
Ida B. Wells & lynching data



Hull-House wage maps



Counter-maps



Louise E. Jefferson,
"Makers of the USA"
(1943)

